ZHENZHONG (JACK) XING

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EDUCATION

Cornell University
Ph.D. in Electrical and Computer Engineering — Expected Summer 2028
M.S. in Electrical and Computer Engineering — Dec 2025 *GPA 4.067/4.3*
Concentrations in Applied Physics and Theory of Computation

Grinnell College
B.A. in Physics (Honors) and Mathematics — May 2022 *GPA 4.0/4.0*

SKILLS

Coding: Cursor, Codex, Claude Code, Python, C, C++, Fortran
Tools: PyTorch, JAX, OpenCV, NumPy, SciPy, Git, Hugging Face
Hardware: PCB design using Altium and EAGLE; analog and digital electronics (MCUs, DACs, and FPGAs); photonics design and layout with Tidy3D and GDSFactory; precision-optics design, assembly, testing, and control; machining and mechanical-system design using Inventor and Fusion 360

SELECTED COURSES

Robotics and ECE: Robot Learning, Robot Control and Optimization, Embedded Systems, Microcontrollers, Random Signals and Processing
Optoelectronics and Physics: Nonlinear and Quantum Optics, Photonics, Electrodynamics, Semiconductor Physics, Quantum Mechanics, Atomic Physics, Quantum Information Theory
Mathematics and CS: Linear and Abstract Algebra, Real and Complex Analysis, Statistical Distance, Matrix Computation, Machine Learning for Science

EXPERIENCE

Cornell University
Postbac research assistant, graduate research student — Jun 2022 - Present

Robotics research

- **Designed and implemented compressed-memory retrieval for VLA steering**. Manuscript in preparation.
  - Compressed the frozen SmolVLA VLM's 960-dimensional output (1.3 KB) into 64-dimensional keys (128 bytes) using a model trained with a soft dynamic time-warping contrastive loss based on action similarity.
  - Initial tests showed significant gains on precision tasks: with a 300-episode memory (1.5 MB), direct rollout success increased from 33% to 62% across 100 held-out peg-insertion tasks.
- **Built and deployed a tabletop robotic arm to automate heterogeneous precision operations in an optics laboratory**. Work submitted to ICRA 2027.
  - Developed and deployed robot perception, control, and motion software that enables operation in the existing lab environment with one additional top-view camera.
  - Developed a multi-camera perception-and-control system with $\sim$0.95 mm localization uncertainty and a task-specific accuracy of $\leq$0.2 mm, enabling reliable automation of sensor manipulation, power measurement, and photonic-chip packaging.

Quantum science research

- **Co-optimized ion-trap junction electrodes and transport trajectories using a Bayesian neural network**. Manuscript in preparation.
  - Applied a BLIPS-style neural network and the cross-entropy method to optimize transport paths and junction designs that reduced the final phonon number by $\sim$100$\times$ for high-speed transport under noise.
  - Built a quadratic-programming solver for dynamic multi-voltage control, reducing ion-transport complexity from $\mathcal{O}(nm)$ to $\mathcal{O}(n)$. Used the resulting voltage solutions to demonstrate ion transport on chips with multilayer integrated photonics for the first time.
- **Demonstrated novel laser-cooling schemes for trapped ions** that reduced total cooling time by $>$10$\times$, addressing a bottleneck in industrial trapped-ion quantum computing systems in which cooling dominates operating time. Results led to two first-author papers.

- **Designed and built a trapped-ion quantum computing system** as a founding student in the Photonics and Quantum Electronics lab.
  - Built a system comprising multistage cryogenic vacuum chambers; DC and RF filtering and stabilization circuits; and free-space optics, fiber optics, and integrated photonics for addressing ions with narrow-linewidth UV–IR lasers. Developed FPGA-, DDS-, and DAC-based electrical and optical controls.
  - Developed and deployed Python packages to accelerate ion-trap chip design and automate system calibration, data collection, and in situ analysis. Several are now used by research groups at ETH and Duke.

Fermilab
Research intern Summer 2021
- Optimized the performance of a gaseous-argon time projection chamber for neutrino detection using Cython; results were published in internal reports.

Grinnell College
Research assistant Summer 2020
- Developed a formalism for describing quantum particles constrained to 2D manifolds using differential geometry. Results were published in *Physica Scripta*.

PUBLICATIONS

Z. Xing, H. M. Rivy. "Robotic Automation of Heterogeneous Precision Operations in an Optics Laboratory." *ICRA* 2027 (submitted).

Z. Xing, H. M. Rivy, V. Natarajan, A. K. Kolhatkar, G. Beck, and K. K. Mehta. "Rapid multi-mode trapped-ion laser cooling in a phase-stable standing wave." *PRX Quantum* (accepted).

A. M. Kolhatkar, V. Natarajan, H. M. Rivy, **Z. Xing**, G. Beck, K. K. Mehta. "Enhanced trapped-ion quantum control in foundry-fabricated integrated devices." *Quantum Information Science, Sensing, and Computation XVIII*, July 2026.

Z. Xing, K. K. Mehta. "Trapped-Ion Laser Cooling in Structured Light Fields." *Physical Review Applied*, July 2025.

S. Rodriguez, L. Rodriguez, **Z. Xing**, C. McMillin, L. R. Ram-Mohan. "Geometrodynamics of a 2D Curved Surface Due to a Constrained Quantum Particle via Its Gravitational Dual: S2 Analytical Model Calculations." *Physica Scripta*, May 2025.

SELECTED AWARDS

Engineering School Fellowship (Cornell University)
Andrew W. Archibald Prize for Highest Scholarship (Grinnell College)
H. George Apostle Prize in Physics (Grinnell College)

LANGUAGES

English (Fluent)
Mandarin (Native)